# Supply Chain data analysis

**Domain knowledge:**

RAW MATERIALS → MANUFACTURING → WAREHOUSE → DISTRIBUTION → CUSTOMER

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┌──────────────────────────────────────────────────────────────────────────┐
│                        SUPPLY CHAIN WORKFLOW                             │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│  ┌──────────┐    ┌──────────────┐    ┌─────────┐    ┌──────────┐    ┌──────┐│
│  │ SUPPLIER │──►│ MANUFACTURING│──►│WAREHOUSE│──►│ SHIPPING │──►│CUSTOM││
│  └──────────┘    └──────────────┘    └─────────┘    └──────────┘    └──────┘│
│      │               │               │              │             │   │
│      ▼               ▼               ▼              ▼             ▼   │
│ Supplier name    Production      Stock levels    Shipping       Revenue │
│ Location         volumes         Availability    carriers       Customer│
│ Supplier_Lead    Manufacturing   Price           Shipping times  demographics
│ _Time            costs           Order_Lead      Carrier_
│                  Inspection      _Time           Shipping_Fee
│                  results         Order           Transportation
│                  Defect rates    quantities      modes
│                                    Number sold    Routes
│                                                   Total_Logistics
│                                                   _Cost
└──────────────────────────────────────────────────────────────────────────┘
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RESEARCH QUESTIONS

**Question 1: One-Way ANOVA**

- Research Question: **Does the Product Type significantly affect the Manufacturing Costs?**
- Hypothesis: * *Null :* The mean manufacturing cost is the same for Skincare, Haircare, and Cosmetics.
    - *Alternative :* At least one product type has a different mean manufacturing cost.

**Question 2: Two-Way ANOVA (With Interaction)**

- Research Question: **How do Shipping Carrier and Location interact to affect Shipping Costs?**
- Hypothesis: * *Factor A:* Shipping costs differ by Carrier.
    - *Factor B:* Shipping costs differ by Location.

- *Interaction:* The effect of the Carrier on cost depends on the Location (e.g., Carrier A might be cheaper in Mumbai but more expensive in Kolkata).

**3. One-Way ANOVA (Operations Focus)**
**Research Question:** Does mean lead time differ by supplier?

- **Managerial Insight:** Identifies which suppliers are most efficient and helps optimize procurement decisions.

**4. Two-Way ANOVA (With Interaction – Quality & Logistics)**
**Research Question:** Do transportation mode and product type interact to affect defect rates?

- **Main Effects:**
  - Defect rates differ by transportation mode
  - Defect rates differ by product type
- **Interaction:**
  - The effect of transportation mode on defect rates depends on the product type (e.g., fragile products may be more sensitive to certain modes)

## Refined research Questions:

We must provide **at least one appropriate graph and table** per conclusion

### Question 1: One-Way ANOVA (Product & Cost)

- **Research Question:** Does the product category (Skincare, Haircare, or Cosmetics) have a statistically significant effect on the mean manufacturing costs of the items?
- **Null Hypothesis ($H_0$):** The mean manufacturing cost is equal across all three product categories (Skincare, Haircare, and Cosmetics).
- **Alternative Hypothesis ($H_a$):** At least one product category has a mean manufacturing cost that is significantly different from the others.

### Question 2: Two-Way ANOVA with Interaction (Logistics)

- **Research Question:** How do the choice of shipping carrier and the geographic location interact to influence the total shipping costs?

- **Null Hypotheses ($H_0$):** * There is no main effect of the shipping carrier on mean shipping costs.
    - There is no main effect of the geographic location on mean shipping costs.
    - There is no interaction effect between shipping carrier and location on mean shipping costs.
- **Alternative Hypotheses ($H_a$):** * The shipping carrier significantly affects mean shipping costs.
    - The geographic location significantly affects mean shipping costs.
    - A significant interaction exists, meaning the effect of the carrier on cost depends on the specific location.

## Question 3: One-Way ANOVA (Operations)

- **Research Question:** Is there a significant difference in the mean lead times among the various suppliers in the supply chain?
- **Null Hypothesis ($H_0$):** The mean lead time is the same for all suppliers included in the study.
- **Alternative Hypothesis ($H_a$):** At least one supplier has a mean lead time that differs significantly from the rest.

## Question 4: Two-Way ANOVA with Interaction (Quality Control)

- **Research Question:** Do transportation modes and product types interact to significantly impact the observed defect rates of the products?
- **Null Hypotheses ($H_0$):** * The transportation mode has no significant effect on the mean defect rate.
    - The product type has no significant effect on the mean defect rate.
    - There is no interaction between transportation mode and product type regarding defect rates.
- **Alternative Hypotheses ($H_a$):** * The transportation mode significantly impacts the mean defect rate.
    - The product type significantly impacts the mean defect rate.
    - A significant interaction exists, where the impact of the transportation mode on the defect rate varies depending on the product type.

1. Clean Dataset

**Rename 4 columns**

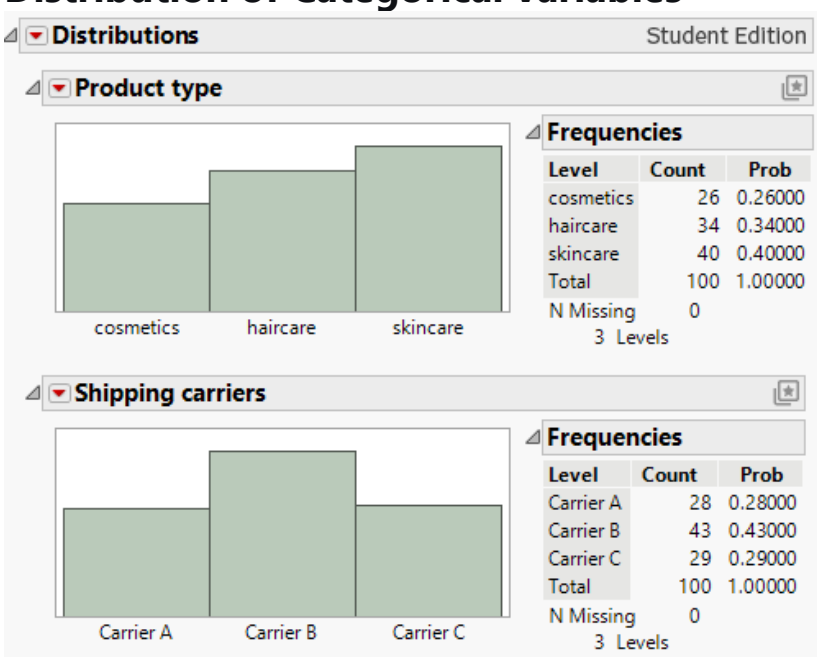
**Shipping costs** Fee paid to shipping carrier only

**Costs (last column)** Total cost = manufacturing + shipping + handling + maybe other operational costs

| Row | Lead times (col 9) | Lead time (col 16) | Manufacturing lead time (col 18) | costs |
|---|---|---|---|---|
| changed | Supplier_Lead_Time | Order_Lead_Time | Manufacturing_Lead_Time | Total_Costs |

2. **Graph Builder**

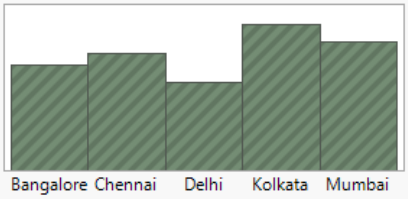
## A. Distribution of Categorical variables



## Distributions

Student Edition

### Location



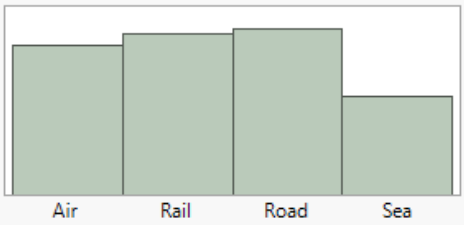

#### Frequencies

| Level | Count | Prob |
|---|---|---|
| Bangalore | 18 | 0.18000 |
| Chennai | 20 | 0.20000 |
| Delhi | 15 | 0.15000 |
| Kolkata | 25 | 0.25000 |
| Mumbai | 22 | 0.22000 |
| Total | 100 | 1.00000 |

N Missing 0

5 Levels

### Transportation modes



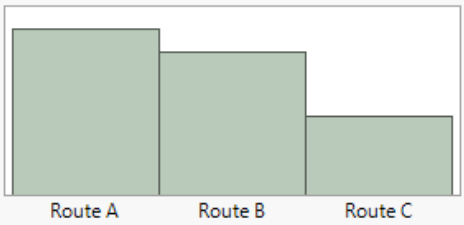

#### Frequencies

| Level | Count | Prob |
|---|---|---|
| Air | 26 | 0.26000 |
| Rail | 28 | 0.28000 |
| Road | 29 | 0.29000 |
| Sea | 17 | 0.17000 |
| Total | 100 | 1.00000 |

N Missing 0

4 Levels

### Routes




#### Frequencies

| Level | Count | Prob |
|---|---|---|
| Route A | 43 | 0.43000 |
| Route B | 37 | 0.37000 |
| Route C | 20 | 0.20000 |
| Total | 100 | 1.00000 |

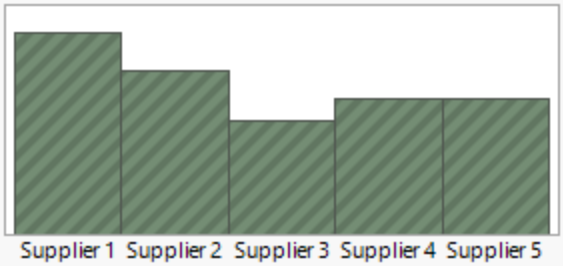
N Missing 0

3 Levels

## Distributions

Student Edition

### Supplier name




#### Frequencies

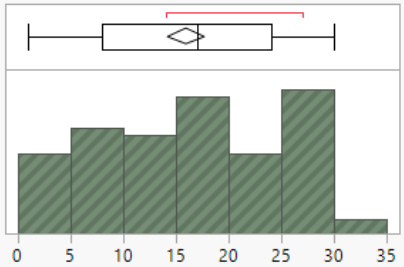
| Level | Count | Prob |
|---|---|---|
| Supplier 1 | 27 | 0.27000 |
| Supplier 2 | 22 | 0.22000 |
| Supplier 3 | 15 | 0.15000 |
| Supplier 4 | 18 | 0.18000 |
| Supplier 5 | 18 | 0.18000 |
| Total | 100 | 1.00000 |

N Missing 0

5 Levels

# B. Distribution of Numerical Variables

**Supplier_Lead_Time**




**Quantiles**

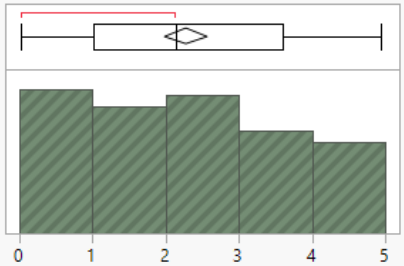
| | | |
|---|---|---|
| 100.0% | maximum | 30 |
| 99.5% | | 30 |
| 97.5% | | 29.475 |
| 90.0% | | 27 |
| 75.0% | quartile | 24 |
| 50.0% | median | 17 |
| 25.0% | quartile | 8 |
| 10.0% | | 3.1 |
| 2.5% | | 1 |
| 0.5% | | 1 |
| 0.0% | minimum | 1 |

**Summary Statistics**

| | |
|---|---|
| Mean | 15.96 |
| Std Dev | 8.7858012 |
| Std Err Mean | 0.8785801 |
| Upper 95% Mean | 17.703294 |
| Lower 95% Mean | 14.216706 |
| N | 100 |
| N Missing | 0 |

JMP Only

**Defect rates**



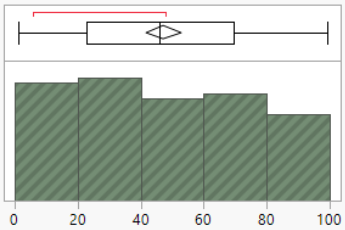

**Quantiles**

| | | |
|---|---|---|
| 100.0% | maximum | 4.9392553 |
| 99.5% | | 4.9392553 |
| 97.5% | | 4.8811563 |
| 90.0% | | 4.3538651 |
| 75.0% | quartile | 3.609894 |
| 50.0% | median | 2.1418627 |
| 25.0% | quartile | 1.0038237 |
| 10.0% | | 0.3346914 |
| 2.5% | | 0.0338394 |
| 0.5% | | 0.0186076 |
| 0.0% | minimum | 0.0186076 |

**Summary Statistics**

| | |
|---|---|
| Mean | 2.277158 |
| Std Dev | 1.4613655 |
| Std Err Mean | 0.1461366 |
| Upper 95% Mean | 2.5671246 |
| Lower 95% Mean | 1.9871914 |
| N | 100 |
| N Missing | 0 |

**Manufacturing costs**



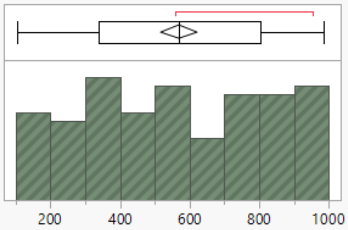

**Quantiles**

| | | |
|---|---|---|
| 100.0% | maximum | 99.466109 |
| 99.5% | | 99.466109 |
| 97.5% | | 98.199981 |
| 90.0% | | 91.676585 |
| 75.0% | quartile | 69.493239 |
| 50.0% | median | 45.905622 |
| 25.0% | quartile | 22.697171 |
| 10.0% | | 9.0219116 |
| 2.5% | | 1.756581 |
| 0.5% | | 1.0850686 |
| 0.0% | minimum | 1.0850686 |

**Summary Statistics**

| | |
|---|---|
| Mean | 47.266693 |
| Std Dev | 28.982841 |
| Std Err Mean | 2.8982841 |
| Upper 95% Mean | 53.017518 |
| Lower 95% Mean | 41.515869 |
| N | 100 |
| N Missing | 0 |

**Production volumes**




**Quantiles**

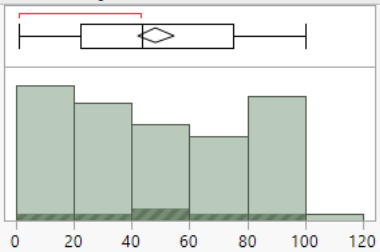
| | | |
|---|---|---|
| 100.0% | maximum | 985 |
| 99.5% | | 985 |
| 97.5% | | 967.325 |
| 90.0% | | 920.8 |
| 75.0% | quartile | 803 |
| 50.0% | median | 568.5 |
| 25.0% | quartile | 340 |
| 10.0% | | 198.4 |
| 2.5% | | 131.575 |
| 0.5% | | 104 |
| 0.0% | minimum | 104 |

**Summary Statistics**

| | |
|---|---|
| Mean | 567.84 |
| Std Dev | 263.04686 |
| Std Err Mean | 26.304686 |
| Upper 95% Mean | 620.0342 |
| Lower 95% Mean | 515.6458 |
| N | 100 |
| N Missing | 0 |

**Distributions** Student Edition

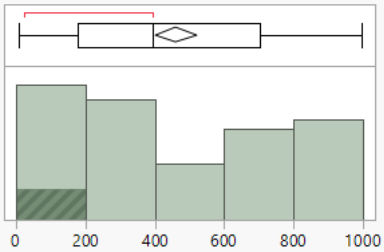
**Availability**




| Quantiles | | |
|---|---|---|
| 100.0% | maximum | 100 |
| 99.5% | | 100 |
| 97.5% | | 98.475 |
| 90.0% | | 93 |
| 75.0% | quartile | 75 |
| 50.0% | median | 43.5 |
| 25.0% | quartile | 22.25 |
| 10.0% | | 10.1 |
| 2.5% | | 2.05 |
| 0.5% | | 1 |
| 0.0% | minimum | 1 |

| Summary Statistics | |
|---|---|
| Mean | 48.4 |
| Std Dev | 30.743317 |
| Std Err Mean | 3.0743317 |
| Upper 95% Mean | 54.500141 |
| Lower 95% Mean | 42.299859 |
| N | 100 |
| N Missing | 0 |

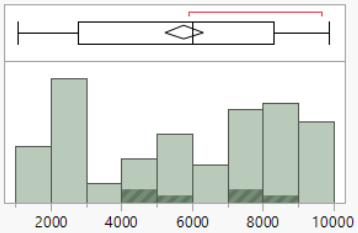
**Number of products sold**




| Quantiles | | |
|---|---|---|
| 100.0% | maximum | 996 |
| 99.5% | | 996 |
| 97.5% | | 983.325 |
| 90.0% | | 915.7 |
| 75.0% | quartile | 704.75 |
| 50.0% | median | 392.5 |
| 25.0% | quartile | 178.75 |
| 10.0% | | 93.6 |
| 2.5% | | 24.525 |
| 0.5% | | 8 |
| 0.0% | minimum | 8 |

| Summary Statistics | |
|---|---|
| Mean | 460.99 |
| Std Dev | 303.78007 |
| Std Err Mean | 30.378007 |
| Upper 95% Mean | 521.26656 |
| Lower 95% Mean | 400.71344 |
| N | 100 |
| N Missing | 0 |

**Revenue generated**




| Quantiles | | |
|---|---|---|
| 100.0% | maximum | 9866.4655 |
| 99.5% | | 9866.4655 |
| 97.5% | | 9672.797 |
| 90.0% | | 9346.7247 |
| 75.0% | quartile | 8297.2611 |
| 50.0% | median | 6006.352 |
| 25.0% | quartile | 2781.844 |
| 10.0% | | 2023.8638 |
| 2.5% | | 1262.5964 |
| 0.5% | | 1061.6185 |
| 0.0% | minimum | 1061.6185 |

| Summary Statistics | |
|---|---|
| Mean | 5776.0482 |
| Std Dev | 2732.8417 |
| Std Err Mean | 273.28417 |
| Upper 95% Mean | 6318.3033 |
| Lower 95% Mean | 5233.7931 |
| N | 100 |
| N Missing | 0 |

1. **Which product type generates the highest revenue?**
   → Graph Builder: Product type vs Revenue (box plot or bar chart)

2. **Do higher defect rates reduce revenue?**
   → Scatter plot: Defect rates vs Revenue generated

3. **Which shipping carrier is cheapest and most reliable?**
   → Tabulate: mean(Shipping costs) and mean(Defect rates) by Shipping carriers

4. **Does location (Supplier name, City) affect lead times or costs?**
   → Graph Builder: Location vs Lead time + Manufacturing lead time
5. **How does Customer demographics relate to product type or revenue?**
   → Mosaic plot in Graph Builder